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DWG. NO.	REV.	SHT. 1

REVISION	DESCRIPTION	DATE	APPROVED

AG5800 Thermal Considerations

AG5800 efficiency is 89% efficient. At 80W this would be 8.8W of heat loss

2.4.4 Output Power

The maximum available output power of the Ag5800 is limited by the available power supplied to the module, as a result, when powered by a DC supply the Ag5800 will be able to supply the rated 100W*. However, when used with an IEEE802.3bt PSE, which can only provide a maximum of 89W from its output, then the maximum achievable output power will be 85W.

* When the output voltage is adjusted to 8v maximum the power must be limited to 100W, e.g. with V_{out} set to 20V the output current must be reduced to 5.4A.

If the Ag5800 is thermally connected to the PCB only, then 85W can only be obtained up to 55°C as shown in Figure 13.

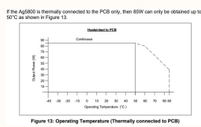


Figure 13: Operating Temperature (Thermally connected to PCB)

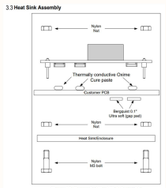
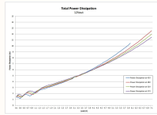


Figure 16: Suggested Heat Sink Assembly



24V
short J3 2 and 3

12V
short J3 1 and 2
short J4 1 and 2

Ferrite beads. 0.1 DC resistance. 2x3x0.1x1.8x1.8= 2W loss

Each diode should see about 1A max which should be about 0.5V Vf. That is about 0.5W dissipation. Across 4 diodes = 2W loss

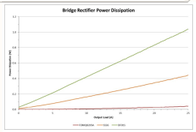


Figure 2: Bridge rectifier dissipation

HV

smaller FB alt: BLM31KN102SN1L

D1 SS36, D2 SS36, D3 SS36, D4 SS36, D6 SMAJ58A, D7 SS36, D8 SS36, D9 SS36, D10 SS36

Vf = 700mV @ 3A

FB1 1k, FB2 1k, R21 0, R22 0

IEEE802.3bt 85W PD

U1 AG5800: THERM, VIN+, VIN-, PoH_En, TYP4-Det, TYP3-Det, AT-DET, SA2, SA1, VOUT2, VOUT2, 0V2, 0V2, VOUT1, VOUT1, 0V1, 0V1, ADJ

THERM_ISO, VIN2+, VIN2- (R22 0 DNP), SA2, SA1

C5 1000uF 35V, C6 1000uF 35V, C7 4.7nF 2kV, C8 4.7nF 2kV

J3, J4, TP1, TP11, C21 100nF, C2 100nF, R3 0, TP2

U2 LM74700QDBVTQ1: ANODE, GATE, CATHODE, VCAP, EN, GND

Q1 HSBB6066 60V 5.2mOhm 60A, C4 100nF

R9 91k, R20 8.2k DNP

ADJ
Open: 12V, 24V
91k to VOUT1: 10.5V, 21V
8.2k to 0V1: 15V, 30.2V

Power Output

VOUT, TP6, TP7, C1 150uF, J1 DB125-3.5-4P-GN-S

PoE-Type LED

U3 OR-3H7B-TP-G, U4 OR-3H7B-TP-G, U6 OR-3H7B-TP-G

Type2, Type2-Det, Type3, Type3-Det, Type4, Type4-Det

R5 2.2k, R8 2.2k, R6 2.2k, D12 (BOTTOM, TOP), VOUT1

Optional DC+ LED

DC+, R7 2.2k, J5 PZ254V-11-02P DNP

Barrel Jack input

CONN3 PJ-050AH, TP4, TP8, DC+, D5 SMAJ30A, C22 100nF, C19 100nF, R23 0, TP3

U5 LM74700QDBVTQ1: ANODE, GATE, CATHODE, VCAP, EN, GND

Q2 HSBB6066 60V 5.2mOhm 60A, C20 100nF

Power + Data Input

CONN1 R-RJ45R08P-C000: Green, Yellow, VIN2-, PWR_LED_ISO, LINK_LED_ISO, SHIELD, SHIELD

Data Passthrough

T1 H2415G: X1+, CX1, X1-, X2+, CX2, X2-, X3+, CX3, X3-, X4+, CX4, X4-, D1+, CD1, D1-, D2+, CD2, D2-, D3+, CD3, D3-, D4+, CD4, D4-

TxRxA_P, TxRxA_N, TxRxB_P, TxRxB_N, TxRxC_P, TxRxC_N, TxRxD_P, TxRxD_N — DIFF100A

TxRxA1_P, TxRxA1_N, TxRxB1_P, TxRxB1_N, TxRxC1_P, TxRxC1_N, TxRxD1_P, TxRxD1_N — DIFF100B

CONN2 KH-5224-8P8C: SHIELD, SHIELD

R10 75, R11 75, R12 75, R13 75, R14 75, R15 75, R16 75, R17 75

C9 10nF, C10 10nF, C11 10nF, C12 10nF, C13 10nF, C14 10nF, C15 10nF, C16 10nF

C17 1000pF 2kV, C18 1000pF 2kV, CGND

Alt: Wurth 7490220122
Eval uses 749052050 which is a more expensive 10G 4PPOE

A better configuration would be to eliminate this transformer and just use the 4 PoE taps on the Raspberry Pi. But the RPi only supports IEEE 802.3at (25W).

Auto PoE-PD Shut Off

J6 PZ254V-11-02P, J7 PZ254V-11-02P, SA2, SA1, R24 10k, U9 OR-3H7B-TP-G, R25 2.2k, J2 PZ254V-11-02P, DC+

Short J7: Phihong PD Signature
Short J7, Short J6: PD Disable (Invalid Signature)
J7 and J6 can be shorted using 100mil jumper or using external user switch
Short J2: Auto PD Disable. When DC+ is present it will create an invalid signature.
This prevents idle power dissipation from AG5800

External Pins

J8 PZ254V-11-04P: Type4, Type3, Type2

Optional wiring the Type4/3/2 pins to RPi GPIO to know what type of PoE source is connected

M3 Standoffs

MP1 SMTSOM380BTR, MP2 SMTSOM380BTR, MP3 SMTSOM380BTR DNP, MP4 SMTSOM380BTR, CGND

PD Power LED (RJ45 Green)

U7 OR-3H7B-TP-G, VIN2+, R18 10k, PWR_LED_ISO, PWR_LED, R4 2.2k, VOUT1

ETH Link LED (RJ45 Yellow)

U8 OR-3H7B-TP-G, VIN2+, R19 10k, LINK_LED_ISO, R26 330, RPI_3V3, LINK_LED, J9 1770953

Wire the LINK_LED and RPI_3V3 to Raspberry Pi RJ45 Pins

BEOVOX-POE
v2.0
Josh Horne
Basement Labs LLC

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	
REFERENCE DOCUMENTS	
BOM:	
ASSY DWG:	
FAB DWG:	
PCB DWG:	

PROJECT: Altium

PROJECT REVISION: Not in version control | DOCUMENT REVISION: Not in version control | DESIGN ITEM:

TITLE: BEOVOX-POE

SIZE	CAGE CODE	DWG NO.	REV
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